

PROJECT REPORT

Enterprise Endpoint Detection & Log Pipeline Analysis

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1. Executive Summary

This project demonstrates the successful design, deployment, and validation of an enterprise-grade log ingestion pipeline. The core objective was to simulate a critical administrative tool abuse technique (PowerShell execution policy bypass) on a target endpoint, verify the centralized collection of security event telemetry, and conduct a professional SOC investigation within Splunk to analyze the execution indicators of compromise (IOCs).

1.1 Target MITRE ATT&CK Techniques

- **T1059.001 (PowerShell):** Leveraging script execution capabilities to run malicious payloads.
- **T1562.001 (Disable or Evade Tools):** Utilizing execution policy bypasses to bypass native host restrictions.

2. Lab Infrastructure & Pipeline Topology

To simulate an authentic enterprise security deployment, I configured a host-only virtualized network. This setup segregates security validation activities from the physical network while mirroring corporate logging topology.

```
+-----+ Port 9997 (TCP) +-----+
| WINDOWS 10 VM | | WINDOWS 11 HOST |
| (Target Host / Forwarder) | -----> | (SIEM Host / Collector) |
| | | |
| [Sysmon Operational Event Logs] | | [Splunk Enterprise Instance] |
| [Splunk Universal Forwarder Service] | | [Centralized Indexer & Search Head] |
+-----+ +-----+
```

3. Threat Simulation & Execution

Threat actors frequently compromise target systems by sending phishing emails with malicious attachment links. Once opened, these attachments spawn underlying command interpreters that run with bypassed execution parameters to fetch malicious tools from remote command-and-control (C2) servers.

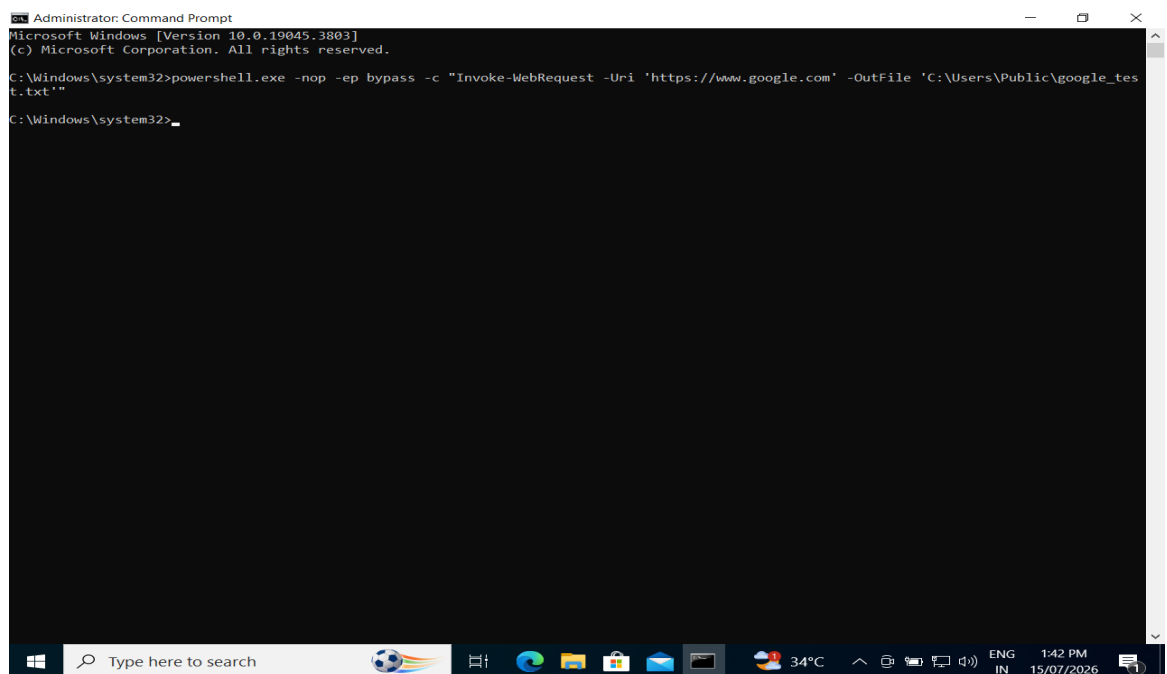
To replicate this specific evasion threat, I executed a PowerShell string engineered to run non-interactively, override default client script restrictions, and silently pull an external text payload down to a publicly writeable directory.

3.1 Simulation Command Line

```
powershell.exe -nop -ep bypass -c "Invoke-WebRequest -Uri 'https://www.google.com' -OutFile 'C:\Users\Public\google_test.txt'"
```

3.2 Endpoint Evidence Capture

Below is the confirmation of the successful execution of the command on the target Windows 10 VM:



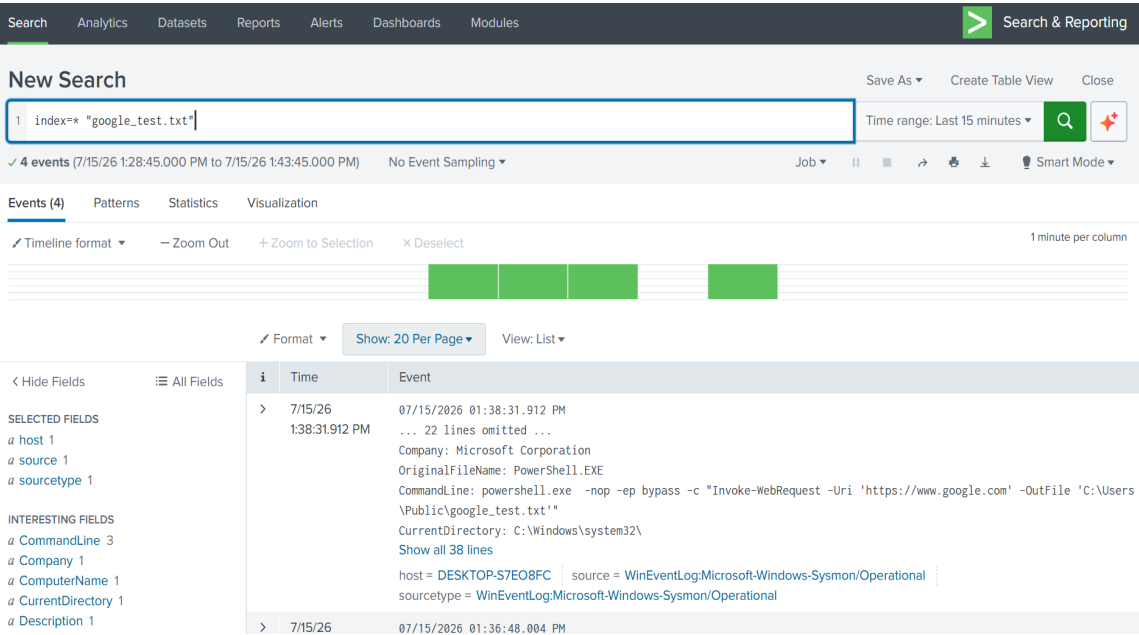
4. SOC Analyst Search & Investigation (Splunk)

Using the Splunk Search and Reporting interface on the Windows 11 host, the indexed logs were queried. To locate the specific action, a high-fidelity search was built targeting the customized download artifact:

```
index=* "google_test.txt"
```

The Splunk search returned 4 unique, high-fidelity security events. By drilling down into the payload, the **Sysmon Event ID 1 (Process Creation)** log was isolated. This event tracks whenever a process is initialized, providing deep forensic properties that native security logs often omit.

4.1 Security Analyst Dashboard Evidence



4.2 Forensic Artifact Deep-Dive

Forensic Field	Discovered Values / Metrics	Security Relevance & Context
host	DESKTOP-S7EO8FC	Confirms the exact endpoint system compromised by the execution.

source	WinEventLog:Microsoft-Windows-Sysmon/Operational	Validates that Sysmon logging was active and captured the process-level activity.
OriginalFileName	PowerShell.EXE	Identifies the true system tool used to download the arbitrary file.
CommandLine	powershell.exe -nop -ep bypass -c "Invoke-WebRequest..."	Discloses the full malicious payload structure, revealing the attempt to bypass security policies.
CurrentDirectory	C:\Windows\system32\	Proves the attacker successfully spawned the payload with highly privileged system context.
ParentImage	cmd.exe	Establishes the parent-child hierarchy, proving PowerShell was spawned from a command console.

5. Incident Response Playbook (L1 Analyst Workflow)

If an identical event were detected in an enterprise production environment, a SOC Level 1 Analyst must execute the following structured playbook:

1. **Alert Triage & Validation:** Categorize this event as High Severity due to the presence of the -ep bypass (Execution Policy Bypass) flag, which strongly indicates intentional defense evasion.

2. **System Isolation (Containment):** Network-isolate host DESKTOP-S7EO8FC immediately using the EDR agent to prevent any lateral movement.
3. **Payload Eradication:** Terminate the process IDs associated with the malicious execution, and safely delete the downloaded file from the C:\Users\Public\ directory.
4. **Host Hardening:** Configure AppLocker parameters on the domain controller to deny execution privileges of PowerShell.exe for non-administrative users.

6. Conclusion

This project demonstrates the powerful visibility achieved by pairing robust endpoint monitors like Sysmon with scalable central indexing engines like Splunk Enterprise. By successfully configuring the pipeline, simulating an attack, and analyzing the resulting artifacts, I have proven my competency in fundamental SOC analyst methodologies required for an enterprise environment.